

NLF: A RESISTOR-NETWORK FRAMEWORK AND LINEAR-TIME SOLVER FOR CONVEX NETWORK-FLOW EQUILIBRIA*

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Abstract. We present **NLF** (Nonlinear Laplacian Flow), a unified framework and linear-time solver for convex network-flow equilibria. Congestion (traffic) routing, minimum-delay communication routing, and maximum flow—each the stationarity of a convex, edge-separable energy—share one form: the *nonlinear graph Laplacian* $B\rho(B^\top\phi) = \alpha d$, where a monotone edge law ρ_e encodes the physics (undirected graphs; directed variants are future work). NLF solves it by an inexact damped chord-Newton iteration whose frozen global linearization—a weighted graph Laplacian—is inverted by a lazily refreshed near-linear Laplacian solver (engine-agnostic; default: approximate Cholesky, LAMG+ interchangeable). The whole nonlinear solve costs 2–4 linear Laplacian solves, so a near-linear inner solve makes the wall-clock empirically $\mathcal{O}(m)$ in the edge count m (not a proved bound). On single-commodity congestion over real road-network topologies (BPR cost), NLF converges for all 2,003 SuiteSparse corpus graphs up to 1.8×10^7 edges and on three larger graphs up to 5.6×10^7 edges. Against a state-of-the-art interior-point method (IPM), NLF was a median $2.6\times$ faster where both converge and $> 45\times$ on the poorly-separable graphs where the IPM’s direct KKT core is super-linear; against matrix-free L-BFGS, a median $4.2\times$ faster and the only solver to finish—within a $5\times$ wall-clock budget—on the 90 hardest instances. A multicommodity extension routes K commodities through one shared hierarchy at $\mathcal{O}(Km)$ per step; on real capacity data, congestion inflates shared-corridor costs (up to $8.8\times$) without rerouting. The same machinery recovers the exact combinatorial max-flow as a short sequence of Laplacian solves, with the cut potential as a by-product. Code and benchmarks: <https://github.com/orenlivne/nlf>.

Key words. network equilibrium, traffic assignment, maximum flow, algebraic multigrid, nonlinear multigrid, inexact Newton methods, continuation, graph Laplacian

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1. Introduction. Flows on networks at equilibrium are everywhere: drivers distributing over a road map until no route is faster (traffic assignment [9, 10, 11]), packets routed to minimize delay in a communication network [15, 16], current in a resistor network [26], and—at the combinatorial extreme—the maximum flow through a capacitated graph [17, 18, 20]. These look like different problems with different algorithms: traffic assignment is solved by Frank–Wolfe or general convex optimization, max-flow by augmenting-path or push–relabel combinatorics. We make a unifying observation and build a single fast solver on it.

The observation. Each of these is the stationary point of a convex, edge-separable energy in the node potentials, and its Euler equation is one and the same *nonlinear graph Laplacian*,

$$(1.1) \quad B\rho(B^\top\phi) = \alpha d,$$

where B is the node–edge incidence matrix, ϕ the node potentials, d a source/sink demand, α a load, and ρ_e a monotone edge law that alone encodes the physics: a saturating ρ_e gives a hard capacity (max flow); a gently rising ρ_e gives a congestion cost (traffic)¹. The flow is $f_e = \rho_e((B^\top\phi)_e)$, and every Newton linearization of (1.1) is an ordinary weighted graph Laplacian $J = B \operatorname{diag}(\rho') B^\top$. For congestion and routing problems α is a given input; for maximum flow, α is an unknown to be maximized.

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¹Notation: ρ acts componentwise and ρ_e denotes the law on an individual edge e .

To obtain a fast *nonlinear* solver, we use an inexact chord-Newton iteration with the linearization frozen across most steps, lazily refreshed. For maximum flow we weave a continuation in α into the same Newton loop to track the solution through the fold. The nonlinear solve costs 2–4 linear Laplacian solves (§4.8). Any near-linear Laplacian solver can serve as the inner engine; we default to approximate Cholesky [25], the faster engine across the corpus at equal robustness (§4.7), with LAMG+ [8] interchangeable in robustness (its constant differs by graph class). One incidental convenience of LAMG+’s affinity-based aggregation is that it naturally respects the saturating min cut, though the framework does not depend on it (§3.1).

1.1. Contribution. We make three contributions, in order of practical weight.

1. *A unifying resistor-network formulation* (§2) that casts maximum flow, congestion (traffic) routing, and minimum-delay routing as one nonlinear Laplacian (1.1), with the problem class fixed by a single monotone edge law and the easy/hard dichotomy (no-fold vs. fold; §2) fixed by whether that law saturates. The stationarity equation itself is not new—it is the optimality system of a convex, edge-separable network-flow program (monotropic programming [43]); our contribution is the unification, that one edge law parametrizes all three problems, and a single solver that addresses the whole class. Unlike the electrical-flow and IPM Laplacian paradigm for flow [23, 24], which drive an outer reweighting/barrier loop around linear solves to an $(1 + \epsilon)$ -approximate optimum, here the nonlinearity sits in the edge law and the equilibrium reached is exact.
2. *A linear-time solver for convex congestion flow* (§4)—establishing two claims that serve as a building block for a larger congestion solver: (a) the convex Beckmann equilibrium is exactly (1.1); (b) a robust $\mathcal{O}(m)$ near-linear Laplacian inner solve makes it linear-time as well. No combinatorial algorithm addresses this problem. NLF matches a state-of-the-art IPM to 10^{-10} in a flat step count; on poorly-separable graphs NLF is the only $\mathcal{O}(m)$ solver and wins past $45\times$ (§4.4). Robustness is corpus-wide: 2,003 SuiteSparse graphs converge in time $\propto m^{1.02}$ (§4.5, 4.8). A multicommodity extension (§6) carries K commodities through one hierarchy at $\mathcal{O}(Km)$. We find that on real capacity data coupling inflates shared-corridor costs ($\leq 8.8\times$) without rerouting.
3. *An $\mathcal{O}(m)$ maximum-flow solver* (§5). The saturating instance recovers the *exact* combinatorial max-flow as a short, bounded sequence of Laplacian solves, with the cut respecting multigrid coarsening arising for free. Not faster than a tuned combinatorial solver, but returns the cut potential and full parametric flow curve.

Extensions are outlined in §7: directed multicommodity assignment, a full-multigrid variant, and related optimization problems: optimal transport and DC optimal flow.

1.2. Motivating applications. Computing an equilibrium flow on a large network is a recurring industrial task, and in each domain the computational bottleneck is the same Laplacian solve at the core of (1.1). The production formulations of these tasks are typically *directed* (arc-based, with asymmetric per-direction costs); this paper studies the *undirected* relaxation as a feasibility study, leaving the directed models these domains deploy to future work (§7). The applications below are thus motivating problem classes; we don’t claim that NLF replaces an incumbent in its directed production setting.

Road traffic assignment. Metropolitan planning agencies compute user-equilibrium traffic assignment, i.e., how drivers distribute over a road map until no route is faster, to evaluate road projects and pricing [9, 10, 11]. Its undirected convex relaxation is the Beckmann program with the BPR congestion law, which is an NLF

no-fold instance and the setting of our sharpest result (§4). The incumbent solvers are Frank–Wolfe and bush-based methods (§4.2).

Communication and data-center routing. Routing packets to minimize end-to-end delay under the Kleinrock $M/M/1$ model [15, 16] is the same equilibrium with a capacity-saturating delay law—a fold instance handled by the continuation of §3.3.

Maximum flow (logistics, scheduling, vision). The capacitated maximum-flow problem [17, 18] is the saturating-law extreme of the same energy (§5). NLF recovers the exact value together with the min cut and the full flow-vs-load curve. Combinatorial solvers are faster to obtain the maximum flow value, but do not return the smooth interior flow or the parametric curve that an outer optimizer manipulates when max flow is an inner step.

Nonlinear resistor and analog-circuit networks. The most literal instance: monotone two-terminal devices (diodes, varistors, memristors) in DC steady state give (1.1) with ρ_e the device I – V law—the form of DC circuit analysis and in-memory-computing crossbars. As with OPF (§7.5) these topologies are typically structured, where direct solvers are efficient; NLF would have an edge only for large, irregular analog networks.

Across all applications, in addition to a scalar optimum (e.g., the maximum flow value) the equilibrium (flow), its binding-constraint structure (min cut), and its sensitivity to load are also required for design optimization, which is the gap NLF targets.

1.3. Prior work, and how NLF differs. Three lines of work meet at (1.1); NLF sits in the gap none of them fills.

The Laplacian paradigm for flow. Electrical-flow algorithms [23] and IPM for generalized flow [24] solve a sequence of linear weighted-Laplacian systems inside an outer multiplicative-weights or barrier loop, reaching an $(1+\epsilon)$ -approximate optimum; the flow is a linear function of the potential and all nonlinearity lives in the outer schedule. NLF instead places the nonlinearity in the constitutive law $f = \rho(B^\top \phi)$, so a single energy encodes the capacity and the equilibrium reached is exact (§2).

Fast linear Laplacian solvers. Nearly-linear solvers for $Lx = b$ such as combinatorial multigrid [6], support-tree/Spielman–Teng theory [5], approximate Cholesky [25, 36], and algebraic multigrid including LAMG [1] address the linear, unconstrained problem only. They are the engine, not the method: NLF calls a robust $\mathcal{O}(m)$ Laplacian solver as its inner kernel but adds the nonlinear, capacity-constrained equilibrium layer on top. AMG has also been applied to the complex-valued non-separable AC power-flow equations [7]—a nonlinear system with no objective and a non-Laplacian Jacobian, solved by nonlinear FAS multigrid with a multiplicative coarse correction (no Newton, no fold); a different problem class from the real, edge-separable, SPD-Laplacian constrained flow program here.

Multilevel optimization. NLF is not multilevel optimization in the MG/OPT sense [29, 30, 31]: it does not build coarse optimization models or recurse the objective. While the equilibrium is nonlinear equation (1.1), NLF employs a near-linear inner solve only as the linear inner solve inside a frozen-linearization chord-Newton continuation; the nonlinearity is carried by the edge law and the outer continuation, not by a coarse-grid objective. However, Sec. 7.5 outlines how NLF might be extended to a multilevel optimization paradigm when a cost functional is added.

Combinatorial and transport-specific solvers. Augmenting-path and push–relabel max-flow [18, 21] and the almost-linear-time exact algorithm [20]—a theoretical breakthrough with galactic constants, not a practical implementation—target directed max/min-cost flow; Frank–Wolfe [13] and bush-based assignment (Algorithm B [38],

TAPAS [39]) target the smooth-cost traffic program. Each is specialized to one problem class. NLF's contribution is orthogonal: a *single* solver, parametrized by one edge law, that spans all of them and inherits graph-class robustness from its multi-grid core. We benchmark against representatives of each family where a runnable one exists (Ipopt [14] for congestion, Boykov–Kolmogorov for max flow) and position the rest honestly where it does not (Frank–Wolfe, bush methods; §4.2).

2. A unified resistor-network formulation. Let B be the $n \times m$ signed node–edge incidence matrix of a connected graph ($B_{ie} = +1$ if node i is the head of edge e , -1 if the tail). Write $\phi \in \mathbb{R}^n$ for the node potentials, $d \in \mathbb{R}^n$ for a balanced demand ($\mathbf{1}^\top d = 0$), and $\alpha \geq 0$ for a scalar load. Each edge e carries a flow f_e driven by its potential difference $g_e = (B^\top \phi)_e = \phi_i - \phi_j$ where i, j are the head and tail of e . The physics of the medium is encoded in a smooth, strictly monotone edge law $\rho_e : \mathbb{R} \rightarrow \mathbb{R}$:

$$(2.1) \quad f_e = \rho_e(g_e).$$

Different choices of ρ_e give different physical problems (Table 2.1): an Ohmic resistor has $\rho_e(g) = g/R_e$; a saturating capacity has $\rho_e(g) = c_e \tanh(g/c_e)$; a BPR congestion link has ρ_e equal to the inverse of the BPR travel-time function.

The equilibrium equation. Flow conservation at every node, $Bf = \alpha d$, together with the constitutive relation (2.1), gives the *NLF master equation* (1.1), a nonlinear system in the potentials ϕ alone. This is the equation NLF solves.

2.1. Primal–dual structure and the co-energy. Because each ρ_e is monotone, it is the derivative of a strictly convex *co-energy* Ψ_e : $\rho_e = \Psi'_e$. Equation (1.1) is then the stationarity condition $\nabla_\phi E = 0$ of the unconstrained, convex *dual energy*

$$(2.2) \quad E(\phi) = \sum_e \Psi_e((B^\top \phi)_e) - \alpha d^\top \phi,$$

Solving (1.1) is therefore equivalent to minimizing E .

The dual energy E arises naturally from a *primal* convex program in the flow variables f . Each edge has a *primal energy* (cost) $\Phi_e(f_e)$, and the primal problem minimizes total cost subject to flow conservation:

$$(2.3) \quad \min_f \sum_e \Phi_e(f_e) \quad \text{s.t.} \quad Bf = \alpha d.$$

The Lagrangian of (2.3) introduces node potentials ϕ as multipliers on the conservation constraint; the KKT stationarity condition $\Phi'_e(f_e) = (B^\top \phi)_e$ says the marginal cost equals the potential difference, i.e. $t_e(f_e) = g_e$, or $f_e = \rho_e(g_e)$. The *marginal cost* $t_e = \Phi'_e = \rho_e^{-1}$ is thus the inverse edge law. Substituting into $Bf = \alpha d$ recovers (1.1).

The relationship between Φ_e and Ψ_e is the *Legendre–Fenchel duality* [44]: $\Psi_e = \Phi_e^*$, i.e. $\Psi_e(g) = \sup_f [g \cdot f - \Phi_e(f)]$, and $\Psi'_e = (\Phi_e^*)' = (\Phi'_e)^{-1} = \rho_e$. In words: the co-energy Ψ_e is the convex conjugate of the primal cost Φ_e , and its derivative is the edge law ρ_e . The potentials ϕ are sometimes called *co-potentials* in the resistor-network literature [26] because they live in the dual (potential) space, conjugate to the primal (flow) space. Appendix A derives Φ_e and Ψ_e for each application in Table 2.1.

The Newton linearization. Differentiating (1.1) with respect to ϕ gives the Newton (Jacobian) matrix

$$(2.4) \quad J = B \operatorname{diag}(\rho'(B^\top \phi)) B^\top, \quad w_e = \rho'_e(g_e) = \Psi''_e(g_e) = 1/t'_e(f_e) \geq 0,$$

a weighted graph Laplacian with *solution-dependent conductances* w_e . Everything problem-specific lives in ρ_e ; the solver never sees anything but ρ_e , ρ'_e , and J .

TABLE 2.1

Three instances of the resistor-network framework (1.1). The edge law ρ_e (flow vs. potential gradient) is the inverse of the marginal cost $t_e = \Phi'_e$. “Fold” marks a feasibility limit where the conductance $\rho'_e \rightarrow 0$ and J becomes singular.

Problem	Marginal cost $t_e(f)$	Conductance ρ'_e	Fold?	Application
maximum flow [17, 19]	box $ f \leq c_e$ (soft barrier)	$\rightarrow 0$ at cut	yes	vision, matching
min-delay routing [15, 16]	$1/(c_e - f)$ (Kleinrock)	$\rightarrow 0$ at $f \rightarrow c_e$	yes	communication
congestion [9, 10, 11]	$t_e^0(1 + b(f/c_e)^p)$	> 0 (no satn.)	no	road traffic

2.2. The fold dichotomy. The single feature that governs difficulty is whether the edge law *saturates*. Table 2.1 lists three standard instances.

- *Soft congestion (rising ρ , no saturation).* The load α is a given input. The cost rises with flow but imposes no hard cap, so ρ_e is unbounded and the law never *saturates*: $\rho'_e > 0$ at every finite flow (for BPR ρ'_e is large near free flow and decays as f grows, but never reaches 0). Hence J is nonsingular at every equilibrium—no feasibility limit and no fold. Traffic assignment with BPR cost is of this type; on its bounded-flow equilibria J is well-conditioned, so (1.1) is solved directly at the target load, with no continuation (§4).
- *Hard capacity (saturating ρ , conductance $\rightarrow 0$).* α is an unknown to be maximized. The flow cannot exceed a capacity c_e , so ρ_e is bounded and $\rho'_e \rightarrow 0$ as the edge saturates. As α rises to a feasibility limit, the saturating edges form a cut and J *folds*—a continuation limit point where α caps at the min cut F^* and J becomes singular. Maximum flow and minimum-delay communication routing (Kleinrock delay) are of this type; they require arclength continuation through the fold (§3.3).

2.3. The max-flow law in detail. For the saturating instance used in §5, take

$$(2.5) \quad \rho_e(g) = \begin{cases} c_e^+ \tanh(g/c_e^+), & g \geq 0, \\ -c_e^- \tanh(g/c_e^-), & g < 0, \end{cases} \quad \Psi_e(g) = c_e^2 \log \cosh(g/c_e),$$

a soft-capacity barrier: $\Psi_e(g) = \frac{1}{2}g^2$ for $|g| \ll c_e$ (a unit resistor) and grows only linearly, $\Psi_e \rightarrow c_e|g|$, for $|g| \gg c_e$ (a saturated edge), so $\rho_e = \Psi'_e$ clamps the flow smoothly to $-c_e^- \leq f_e \leq c_e^+$. Here $w_e = 1 - (f_e/c_e)^2 \in [0, 1]$ represents “edge openness,” 1 for a slack edge and 0 for a saturated one, and **the forming min-cut is exactly the zero-conductance set**, which is why AMG aggregation driven by a relaxation-based strong-connection measure refuses to coarsen across it (§5). The congestion (BPR) law is given where it is used, in §4.

2.4. Relation to prior Laplacian-paradigm formulations. Solving flow through graph-Laplacian systems is the core of a productive line of work, but NLF’s mechanism is fundamentally different. Electrical-flow methods [23] and their p -Laplacian descendants compute a sequence of *linear* electrical flows $f = \text{diag}(w) B^\top \phi$ —each a fixed weighted-Laplacian solve—and steer an *outer* multiplicative-weights loop to a $(1 + \epsilon)$ -approximate max-flow; IPM for (generalized) flow [24] likewise wrap a Laplacian solve in a barrier path-following loop. NLF instead places the nonlinearity in a *constitutive law*: $f = \rho(B^\top \phi)$ is a saturating function of the gradient, so the single energy of (1.1) encodes the capacity exactly and its feasibility limit is the exact min cut. The outer loop is then a short continuation through one scalar fold, not a reweighting or barrier schedule. The Beckmann congestion instance (2.3) is already known [9]; the novelty is the saturating-resistor reformulation of

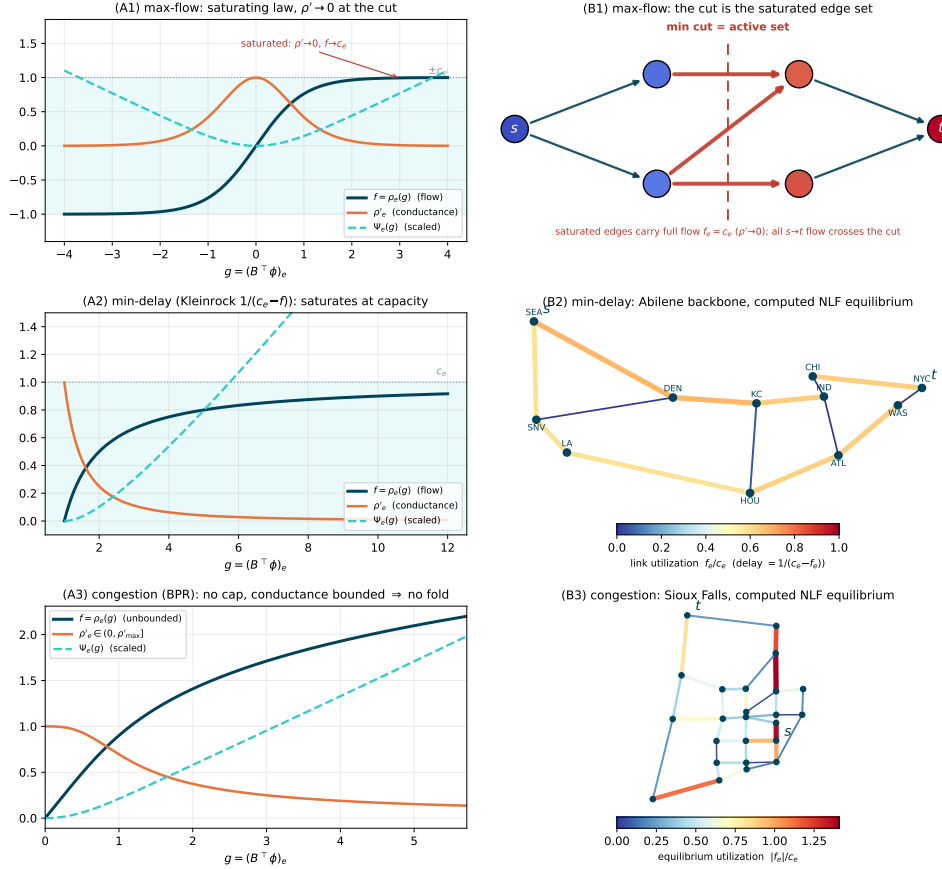


FIG. 2.1. The three instances of the framework (one row per line of Table 2.1); column (A) the edge law ρ_e (flow), its derivative ρ'_e (the solution-dependent conductance) and the energy Ψ_e ; column (B) the behavior it induces. Row 1, max-flow (tanh): the law saturates at $\pm c_e$ and $\rho'_e \rightarrow 0$; (B1) the saturated edges are the min cut = the active set, and ϕ steps across them. Row 2, minimum-delay routing (Kleinrock $1/(c_e - f)$): the law saturates at capacity—the fold class, handled by the continuation of §3.3; (B2) the real application: the Abilene/Internet2 backbone colored by the computed NLF min-delay equilibrium for a Seattle→New York demand. Row 3, congestion (BPR): the law is unbounded and ρ'_e is bounded away from zero—no fold, pure cycling; (B3) the real application: the Sioux Falls road network colored by the computed NLF equilibrium utilization $|f_e|/c_e$ for a single source-sink demand (§4).

max-flow, and placing max-flow, congestion, and minimum-delay routing under one nonlinear Laplacian solved via existing linear Laplacian algorithms.

3. The NLF algorithm. The two problem classes need different outer loops. For congestion (no fold), α is a given input, so (1.1) is solved *directly* at that load by a handful of inexact chord-Newton steps (Algorithm 3.1), warm-started from $\phi = 0$; no load ramp is needed—a single fixed-load solve converges across the corpus (§4.5). For max-flow (fold), α is the unknown maximizer, and continuation from $\alpha = 0$ in pseudo-arclength parametrization tracks the solution curve through the fold to F^* , each fixed- (ϕ, α) subproblem again a handful of chord-Newton steps. The inner solver’s setup—its factor or hierarchy—is built once, frozen across steps and re-formed only

when an observed-convergence monitor signals it has drifted too far. The numbers of steps and inner solves are both bounded, implying empirical- $\mathcal{O}(m)$ complexity. We describe the linear engine (§3.1), the fixed-load solve (§3.2), and the continuation (§3.3).

3.1. The inner Laplacian solve. The outer loop treats the inner solver as a black box that is required to do three things: (i) it solves a graph-Laplacian system $L\phi = b$ in near-linear $\mathcal{O}(m)$ work; (ii) its accuracy is tunable, so it can match the $\eta\|r\|$ forcing tolerance of the chord-Newton step (§3.2) rather than to machine precision; and (iii) its setup is separable from its solve, so the setup can be built once at a frozen linearization and reused across many continuation steps.

Two public near- $\mathcal{O}(m)$ solvers meet all three with different internals. *Approximate Cholesky* [25] sets up a randomized incomplete sparse-Cholesky factor preconditioner and solves by preconditioned conjugate gradients to the requested tolerance. *LAMG+* [8] sets up an aggregation-multigrid hierarchy (relaxation-based affinity aggregation, low-degree Schur elimination, a guarded coarse operator) and solves by a multigrid cycle with min-residual recombination. We use approximate Cholesky as the default as it is the faster default across the corpus at equal robustness (§4.7). LAMG+ is as robust, while its wall-clock constant differs by graph class.

While both inner solvers yield the same overall nonlinear convergence (§4.5), one convenient property of LAMG+’s relaxation-based aggregation is that it is *cut-respecting*: as the cut conductances vanish they decouple relaxation across the cut, so test vectors decorrelate and the affinity declines to aggregate across it. Its measured Asymptotic Convergence Factor (ACF) of ≈ 0.01 thus holds through the continuation with no fold-aware tuning (§5). This is a specific feature of LAMG+, not a framework requirement: adequate refresh of the frozen operator near the fold suffices for any inner solver, and over-freezing degrades all of them equally. (α , the continuation, and the fold never enter the linear solve; the singular cut direction near F^* is deflated outside the engine, §3.3.)

3.2. The fixed-load solve: a frozen-setup inner solve. Eq. (1.1) with fixed α is solved by inexact chord-Newton iteration (Algorithm 3.1). At each step we evaluate the residual $r(\phi) = \alpha d - B\rho(B^\top \phi)$ and solve for the Newton direction $\delta \approx J_H^{-1}r$ using the inner Laplacian solver to tolerance $\eta\|r\|$, $\eta = 0.05$ [33]. The Jacobian J_H and the expensive setup hierarchy are only recalculated when chord Newton’s linear convergence rate rises above 0.25. This addresses the no-fold congestion class; we verified that $\phi = 0$ is always a good initial guess there.

The damped update. The line search (step 5) is standard backtracking on the residual norm [32, Ch. 3]. Because δ comes from a frozen linearization, a full step can overshoot right after a load increase where the law’s curvature is strongest; backtracking by halving and accepting the first non-increasing residual guarantees monotone progress at negligible cost. Near the solution the full step is always accepted (one trial, $\tau = 1$).

When the setup is refreshed. Three conditions trigger: (i) no setup yet; (ii) per-step residual factor > 0.25 (frozen J has drifted); (iii) line-search stall on a stale H —rebuild and retry once; a fresh- H stall breaks the loop. The corpus median refresh count is small (often one). A lazy refresh is 1.5–4.7 $\times$ faster than rebuilding every step.

Deep hierarchies (LAMG+ engine only). On graphs with ≥ 20 -level hierarchies (country-scale roads, large FEM meshes) the dipole right-hand side excites the longest mode that the stock cycle under-treats. A *grown* cycle index (1.15 $\times$ per level, cap

Algorithm 3.1 NLF fixed-load solve

Require: load α ; initial guess ϕ ; inner-solver state H (Cholesky factor or hierarchy, shared across calls); relative residual tolerance tol (default 10^{-9}); forcing parameter η (default 0.05)

- 1: $r \leftarrow \alpha d - B\rho(B^\top \phi)$
- 2: **while** $\|r\| > \text{tol} \cdot \alpha \|d\|$ **do**
- 3: **refresh check:** if H is empty, or the previous pass reduced $\|r\|$ by a factor worse than 0.25, or the line search stalled on a stale H , rebuild $H \leftarrow$ inner setup of $J(\phi) = B \text{diag}(\rho'(B^\top \phi)) B^\top$
- 4: $\delta \leftarrow$ approximate solution of the frozen correction equation $J_H \delta = r$, where $J_H = B \text{diag}(\rho'(B^\top \phi_{\text{build}})) B^\top$ is the linearization H was built from: inner solve(s) with H starting from $\delta = 0$, stopped when $\|r - J_H \delta\| \leq \eta \|r\|$
- 5: **line search:** for $\tau = 1, \frac{1}{2}, \frac{1}{4}, \dots, \frac{1}{2^{49}}$ find the first τ with $\|\alpha d - B\rho(B^\top(\phi + \tau\delta))\| \leq \|r\|$
- 6: **if** line search succeeded **then** $\phi \leftarrow \phi + \tau\delta$
- 7: **else if** H was stale (built before this iteration) **then** rebuild H from current $J(\phi)$; recompute $\delta \leftarrow$ inner solve of $J_H \delta = r$ from $\delta = 0$; repeat line search
- 8: **if** retry succeeded **then** $\phi \leftarrow \phi + \tau\delta$
- 9: **else break** ▷ fresh- H stall: no improving direction exists; exit
- 10: **end if**
- 11: **else break** ▷ fresh H already in use; exit
- 12: **end if**
- 13: $r \leftarrow \alpha d - B\rho(B^\top \phi)$
- 14: **end while**
- 15: **return** ϕ

$\gamma_l \tau_l \leq 0.95$, each cycle stays $\mathcal{O}(m)$) is used, restarted if the factor exceeds 0.6. This engages on 11 of 2,003 graphs (0.5%) and restores textbook convergence.

3.3. Continuation. Only the saturating/fold class needs continuation. Here α is the unknown maximizer. A fixed- α solve is feasible iff $\alpha < F^*$ (the min-cut value), and as $\alpha \uparrow F^*$ the solution curve *folds*— $\alpha(V)$ flattens onto the asymptote F^* while the potentials grow without bound (Fig. 3.1); at the fold $\kappa(J) \sim (F^* - \alpha)^{-1}$ and the source-sink gap is $V = |\phi_s - \phi_t| \sim -\log(F^* - \alpha)$, so $dV/d\alpha \sim (F^* - \alpha)^{-1} \rightarrow \infty$. Thus α is a bad continuation parameter: the feasible loads crowd at the fold and the step is forced to zero. The well-conditioned variable is the *cut-mode amplitude* $\psi = \chi^\top \phi \sim -\log(F^* - \alpha)$,² χ the (a priori unknown) min-cut indicator. We realize this by pseudo-arclength continuation [27, 28] (Algorithm 3.2), which never forms χ .

The equilibria of (1.1) trace a smooth curve $(\phi(s), \alpha(s))$ as the load is varied; we

²Justification of both asymptotics. Near the fold $\phi = V\chi + O(1)$, χ the source-side indicator, so every cut edge carries gradient $g_e = V + O(1)$. The saturating tail of (2.5) is $\rho_e(g) = c_e \tanh(g/c_e) = c_e - 2c_e e^{-2g/c_e} + O(e^{-4g/c_e})$, and all s - t flow crosses the cut, so $\alpha = \sum_{e \in \text{cut}} \rho_e(g_e) = F^* - 2 \sum_{e \in \text{cut}} c_e e^{-2V/c_e} (1 + o(1))$. Hence $F^* - \alpha \asymp e^{-2V/\bar{c}}$, $\bar{c}$ the largest cut capacity, i.e. $V = \frac{\bar{c}}{2} \log \frac{1}{F^* - \alpha} + O(1)$ and $\psi = \chi^\top \phi \propto V$. The same expansion gives the cut conductances $\rho'_e(g_e) = \text{sech}^2(g_e/c_e) = 4e^{-2V/c_e} (1 + o(1))$, whence $\lambda_{\min}(J) \leq \sum_{\text{cut}} \rho'_e \propto F^* - \alpha$ (here and below $\lambda_{\min}$ is the smallest *nonzero* eigenvalue— J being a graph Laplacian, its zero eigenvalue with constant eigenvector $\mathbf{1}$ is excluded, and $\kappa(J)$ is correspondingly the ratio over the nonzero spectrum; the bound is the Rayleigh quotient at χ , which loads only the cut edges); for a *simple* fold the next eigenvalue stays $\Theta(1)$ (the deflated spectral gap) and $\lambda_{\max}(J) = \Theta(1)$, so the bound is two-sided and $\kappa(J) \sim (F^* - \alpha)^{-1}$.

parametrize it by its own arclength s and write a dot for the derivative along the curve, $(\dot{\cdot}) \equiv d(\cdot)/ds$, so the tangent $(\dot{\phi}, \dot{\alpha})$ is a unit vector, $\|\dot{\phi}\|^2 + \dot{\alpha}^2 = 1$. Differentiating (1.1) along s gives:

$$(3.1) \quad J\dot{\phi} = \dot{\alpha}d \implies \dot{\phi} = J^+d, \quad \dot{\alpha} = 1, \quad \text{then rescale to } \|(\dot{\phi}, \dot{\alpha})\| = 1.$$

As $\alpha \rightarrow F^*$, $J^+d \propto \chi/\lambda_{\min}$ (the pseudoinverse J^+ acts on $\mathbf{1}^\perp$, so $\lambda_{\min}$ is again the smallest nonzero eigenvalue), so the tangent rotates onto the cut mode by itself—stepping in s becomes stepping in ψ , the well-conditioned direction.

Each continuation step is a *predictor* that advances a fixed arclength Δs along the tangent, $(\phi, \alpha) \leftarrow (\phi, \alpha) + \Delta s(\dot{\phi}, \dot{\alpha})$, then a *corrector* that drives two residuals to zero: the *equilibrium residual* $r = \alpha d - B\rho(B^\top \phi)$ of (1.1) (as in Algorithm 3.1) and the *arclength residual*

$$(3.2) \quad N = \dot{\phi}^\top(\phi - \phi_0) + \dot{\alpha}(\alpha - \alpha_0) - \Delta s,$$

how far the point has advanced along the tangent past the previous solution (ϕ_0, α_0) , minus the target Δs . Setting $r = 0$ holds the point on the equilibrium curve; $N = 0$ fixes how far along it (hence “pseudo”-arclength: the exact arclength sphere $\|\delta\phi\|^2 + \delta\alpha^2 = \Delta s^2$ is linearized to its tangent plane $N = 0$.) One Newton step on $(r, N) = 0$ is the bordered linear system

$$(3.3) \quad \begin{bmatrix} J & -d \\ \dot{\phi}^\top & \dot{\alpha} \end{bmatrix} \begin{bmatrix} \delta\phi \\ \delta\alpha \end{bmatrix} = \begin{bmatrix} r \\ -N \end{bmatrix}.$$

Block elimination of $\delta\alpha$ gives

$$(3.4) \quad \delta\alpha = -\frac{\dot{\phi}^\top w + N}{\dot{\alpha} + \dot{\phi}^\top u}, \quad \delta\phi = \delta\alpha u + w,$$

where $u = J^+d$ and $w = J^+r$ are computed by two applications of the inner Laplacian solver. Both $u, w \in \text{range}(J^+) = \chi^\perp$: the near-null cut component of the solution (the χ direction) is pinned by the scalar arclength row (3.2), not by these solves. The border keeps the system nonsingular through the fold ($\kappa = \mathcal{O}(10^3)$ while $\kappa(J) \rightarrow \infty$), so the inner Laplacian solver stays fast (approximate Cholesky PCG and LAMG+cycle ACF both ≈ 0.01).

The refresh policy of §3.2 is woven into the corrector as a staleness safeguard (Algorithm 3.2). Both u and w computations reuse one frozen hierarchy; since the cut conductances drain exponentially in the cut amplitude, that hierarchy ages faster as the fold sharpens, and the monitor rebuilds it more often — about every 4 arclength steps at $0.97F^*$, every 2 at $0.995F^*$. The predictor’s tangent $\dot{\phi} = J^+d$, the single direction most sensitive to J ’s drift, is instead recomputed on a fresh hierarchy once per accepted step. The fixed-load solution remains machine-precise as close to the fold as $\alpha = 0.995F^*$ (a fixed- α equilibrium is well-posed only for $\alpha < F^*$), while the arclength continuation tracks through the fold to the exact F^* without stalling (Table 5.1).

4. Transportation: undirected congestion equilibrium. This is NLF’s sharpest result: on convex congestion flow—which has no combinatorial competitor and whose operator never folds—NLF is an empirically linear-time solver, verified against a state-of-the-art interior-point method on real road-network topologies.

Algorithm 3.2 NLF arclength continuation (max-flow fold).

Require: demand d ; maximize α to find F^* $\triangleright$ congestion needs no continuation:
 Algorithm 3.1 at the target load

- 1: $\phi \leftarrow 0, H \leftarrow \emptyset$ $\triangleright$ trivial solution at $\alpha = 0$
- 2: *Seed a feasible load:* $\alpha_{\text{hi}} \leftarrow \sum_{e \sim s} c_e$ ($\geq F^*$, source-cut bound); set $\alpha \leftarrow 0.1 \alpha_{\text{hi}}$ and halve until Algorithm 3.1 converges (so $\alpha < F^*$)
- 3: *Bracket F^* :* try load $g\alpha$ ($g=1.7$) with Algorithm 3.1; accept ($\alpha \leftarrow g\alpha$) while it converges, stop at the first failure ($g\alpha \geq F^*$)
- 4: refresh H ; tangent $(\dot{\phi}, \dot{\alpha})$ by (3.1); initialize arc-step Δs
- 5: **while** $|\dot{\alpha}| > \text{tol}$ **do** $\triangleright \dot{\alpha} \rightarrow 0$ marks the fold, $\alpha \rightarrow F^*$
- 6: *predictor:* $(\phi, \alpha) += \Delta s (\dot{\phi}, \dot{\alpha})$; *corrector:* iterate (3.3) with line search, H lazily refreshed (Algorithm 3.1)
- 7: on success grow Δs and reorient the tangent (3.1); on a fresh- H failure set $\Delta s \leftarrow \Delta s/2$ and re-predict
- 8: **end while**
- 9: **return** $\phi, F^* = \alpha$

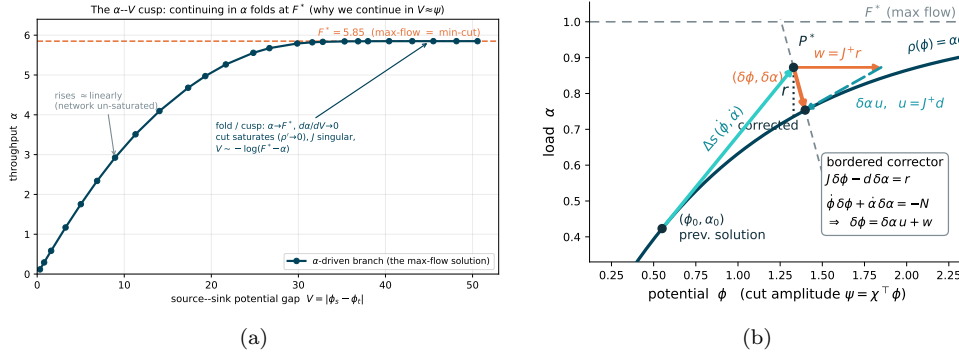


FIG. 3.1. **(a)** The α - V fold (max-flow). As the load α is driven up, the source-sink gap $V = |\phi_s - \phi_t|$ grows; α rises, then folds onto the min-cut value F^* as the cut saturates ($\rho' \rightarrow 0$, J singular) with $V \sim -\log(F^* - \alpha)$. Stepping α crowds the fold; continuing in $V \approx \psi$ keeps steps uniform. Congestion laws have no such fold. **(b)** Pseudo-arclength predictor-corrector, drawn where ϕ, α are both scalars so the equilibria form a plane curve $\rho(\phi) = \alpha d$. From the previous solution the predictor advances Δs along the unit tangent $(\dot{\phi}, \dot{\alpha})$ to an off-curve point P^* ; the corrector drives the equilibrium residual r and the arclength residual N to zero through the bordered system (3.3). Its Newton step splits as $\delta\phi = \delta\alpha u + w$ (block elimination (3.4)) with $u = J^+ d$ along the tangent and $w = J^+ r$ cancelling r ; near the fold $u \propto \chi/\lambda_{\min}$ is large but is pinned by the scalar arclength row, so the inner solves stay well-conditioned.

Scope: an undirected, single-commodity. Throughout this section the congestion problem is the undirected, single-commodity Beckmann relaxation: flows are signed, the cost depends on $|f_e|$, and Wardrop's path-nonnegativity is dropped (App. A). Our experiments establish two claims: (i) this undirected congestion equilibrium is exactly (1.1) for the inverse-BPR law, and (ii) a robust $\mathcal{O}(m)$ linear graph Laplacian inner solve makes it linear-time. The real road networks supply real topologies, capacities, and BPR data exercised with a single source-sink demand; we cross-check NLF against an independent high-accuracy solver on the identical convex program.

4.1. The BPR congestion law. On a congested road the travel time rises with the flow it carries. The field-standard model is the Bureau of Public Roads (BPR)

cost [10],

$$(4.1) \quad t_e(f) = t_e^0 \left(1 + b(f/c_e)^p\right), \quad b = 0.15, \quad p = 4,$$

with t_e^0 the free-flow travel time and c_e the practical capacity. The undirected Beckmann congestion equilibrium (signed flows; Wardrop’s path-nonnegativity relaxed, App. A) is the convex program (2.3) with $\Phi_e(f) = \int_0^f t_e = t_e^0 f + \frac{b t_e^0}{p+1} f^{p+1}/c_e^p$; its edge law is the inverse marginal cost $\rho_e = t_e^{-1}$, with conductance $\rho'_e = 1/t'_e(f) = (t_e^0 b p (f/c_e)^{p-1}/c_e)^{-1} > 0$ at every finite flow: the cost rises but never caps the flow (the law does not saturate), so there is no feasibility limit and no fold.³ The Jacobian $J = B \text{diag}(\rho') B^\top$ is a well-conditioned Laplacian at every bounded-flow iterate, and NLF runs the fixed-load Algorithm 3.1.

4.2. No combinatorial competitor; an interior-point baseline. No combinatorial algorithm addresses the convex congestion program (2.3). Frank–Wolfe [13] converges only sublinearly on the Beckmann program: its $\mathcal{O}(1/k)$ rate (k = iteration count) drives the iteration count to even a 10^{-4} duality gap up with problem size and cannot reach the high accuracy NLF attains—the classical slow-tail limitation of the method on traffic assignment [11]. The field’s high-accuracy standard, the bush-based family (Algorithm B [38], TAPAS [39]), solves the *directed* user equilibrium (nonnegative arc flows, one-sided cost)—a different convex program from the undirected, signed-flow Beckmann relaxation NLF solves here (App. A), so a same-instance head-to-head is ill-posed. We therefore position bush methods qualitatively and benchmark quantitatively against Ipopt [14], a mature IPM solver on the identical program. This is conservative: Ipopt attains the same 10^{-10} , whereas Frank–Wolfe and bush-based methods do not address the max-flow instance (§5). Ipopt’s per-step cost is a sparse-direct KKT factorization: cheap on planar graphs but fill-in grows superlinearly as separators worsen—exactly the axis on which NLF is immune.

4.3. Real road data: NLF and Ipopt. We use the Transportation Networks for Research benchmark [12], the field-standard repository of real metropolitan road-network topologies with real capacities, free-flow times and BPR parameters. Across instances spanning three decades of size, NLF and Ipopt reach the identical equilibrium of the undirected single-commodity Beckmann program (Table 4.1): the Beckmann objective agrees to the solver tolerance and the equilibrium link flows to $\|f_{\text{NLF}} - f_{\text{Ipopt}}\|/\|f_{\text{Ipopt}}\| \sim 10^{-10}$, in ≈ 9 Newton steps with no size trend.

4.4. Linear scaling, and the direct-vs-iterative crossover. Figure 4.1 and Table 4.2 time both solvers on the real road networks (planar, good separators) and on synthetic Erdős–Rényi graphs carrying BPR parameters (poorly separable), to 2.4×10^5 edges. NLF’s per-edge cost is flat on both families—empirically near-linear (the full-corpus fit of §4.5 gives $t \propto m^{1.02}$); neither the step count (≈ 9) nor the inner linear solver ACF sees the graph’s separability, while Ipopt does. On the real road networks its sparse-direct core is near-linear and beats NLF by $\approx 3\times$; planar graphs are where direct methods excel. On poorly-separable graphs the KKT fill-in explodes— $t \propto m^{2.2}$ —so the gap inverts and widens with size: NLF is $5\times$ faster at

³We use a strictly convex, twice-differentiable form of (4.1) with the same free-flow and degree- p congestion structure and real (t_e^0, c_e, b, p) from the data; this regularizes the free-flow threshold so the dual problem is well posed. “Practical capacity” sentinels coded as uncapacitated links are given a large finite capacity.

TABLE 4.1

NLF (BPR congestion, near-linear inner) vs. Ipopt [12] on the undirected, single-commodity Beckmann congestion program over real road-network topologies. Both solvers reach the same equilibrium of this program; the objective matches to the tolerance, link flows to $\sim 10^{-10}$. This is the undirected net-flow relaxation, not the field’s directed origin–destination user equilibrium (§4)—it is an exact cross-check of two solvers on one identical convex program, so its objective value is not comparable to published directed-UE results for these networks. NLF’s step count shows no size trend. (Two further instances, Philadelphia and Berlin-Center, also match to 10^{-10} but need more steps under extreme capacity heterogeneity; omitted.)

network	n	m	NLF obj.	$\ \Delta f\ /\ f\ $	steps
Sioux Falls	24	38	784.96	$3 \cdot 10^{-9}$	7
Anaheim	416	634	1600.92	$6 \cdot 10^{-12}$	9
Chicago-Sketch	933	1 475	8178.62	$3 \cdot 10^{-11}$	9
Austin	7 388	10 591	16615.74	$4 \cdot 10^{-12}$	10
Chicago-regional	12 979	20 627	281.15	$2 \cdot 10^{-11}$	9
Sydney	32 956	38 787	1198.98	$2 \cdot 10^{-11}$	9

TABLE 4.2

Scaling on poorly-separable (Erdős–Rényi) graphs with BPR cost: NLF ($\mathcal{O}(m)$ near-linear inner) vs. Ipopt 3.14.19 (interior-point, MUMPS 5.9.0 sparse-direct KKT). Both reach the same optimum of the undirected congestion program. NLF’s step count shows no size trend; Ipopt is superlinear and exceeds the 120 s CPU budget on the largest instance (“—”). Speed-up = $t_{\text{Ipopt}}/t_{\text{NLF}}$.

m	NLF (s)	steps	Ipopt (s)	speed-up
12 175	0.060	9	0.152	$2.5\times$
29 939	0.173	9	0.856	$4.9\times$
59 732	0.573	9	4.97	$8.7\times$
119 460	1.30	9	58.2	$45\times$
239 842	3.01	9	— (> 120 s)	—

$m = 3.0 \times 10^4$, $45\times$ at $m = 1.2 \times 10^5$, and at $m = 2.4 \times 10^5$ Ipopt does not finish within its 120 s CPU budget while NLF finishes in 3 s.

4.5. Robustness across the graph corpus. The road networks above exercise NLF on real flow topologies; we now show it is robust across graph classes in full. We impose a single-commodity BPR congestion instance (random capacities and free-flow times, a far source–sink demand) on every graph of a 2,003 real-world SuiteSparse collection, each reduced to its largest connected component. This includes spanning structured grids, finite-element and circuit meshes, web, social, road, and citation networks with up to 1.8×10^7 edges.

Under this protocol (one BPR instance per graph, random capacities, a single far source–sink demand, stock settings) NLF converged on all 2,003 graphs to 10^{-9} residual tolerance, with a Newton step count median of 6 and no size trend (Fig. 4.2B). The wall-clock fits $t \propto m^{1.02}$ over the 1,788 graphs above 10^3 edges (Fig. 4.2A), per-edge cost flat at $\approx 0.86 \mu\text{s}/\text{edge}$ median (95th percentile $3.7 \mu\text{s}/\text{edge}$). Three giants confirm the scaling beyond the sweep’s cap, to hollywood-2009 at 5.6×10^7 edges.

4.6. Comparison with other convex optimizers.. We compare three solver paradigms for the same convex BPR program: near-linear (NLF), sparse-direct (Ipopt), and matrix-free first-order (L-BFGS on the dual (2.2); on well-conditioned graphs L-BFGS also scales near-linearly and beats Ipopt, so that win is a fill-in effect any matrix-free method inherits). We ran all three on the 1,669 corpus graphs with

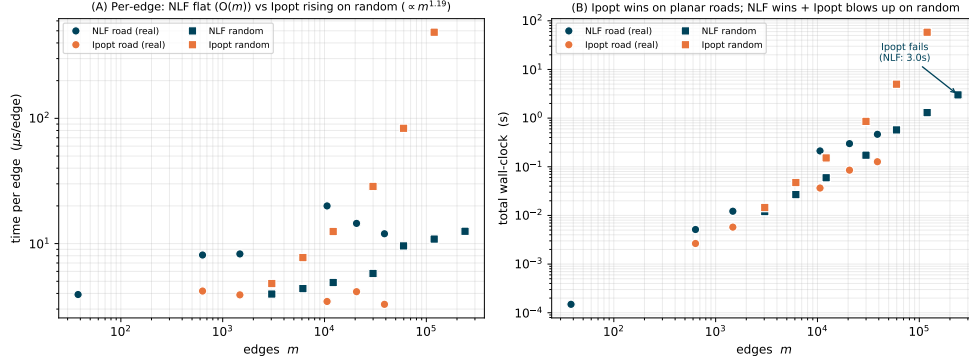


FIG. 4.1. Congestion (BPR) equilibrium, NLF vs. Ipopt, on real road networks (circles) and synthetic poorly-separable graphs (squares). (A) Per-edge time: NLF is flat (near- $\mathcal{O}(m)$) on both families; Ipopt is flat on roads but rises steeply on random graphs ($\propto m^{1.2}$ per edge). (B) Total wall-clock: Ipopt wins on planar roads (cheap separators), is overtaken on random graphs by a margin that widens to 45 $\times$, and fails (time limit) at $m = 2.4 \times 10^5$ where NLF finishes in 3.0 s.

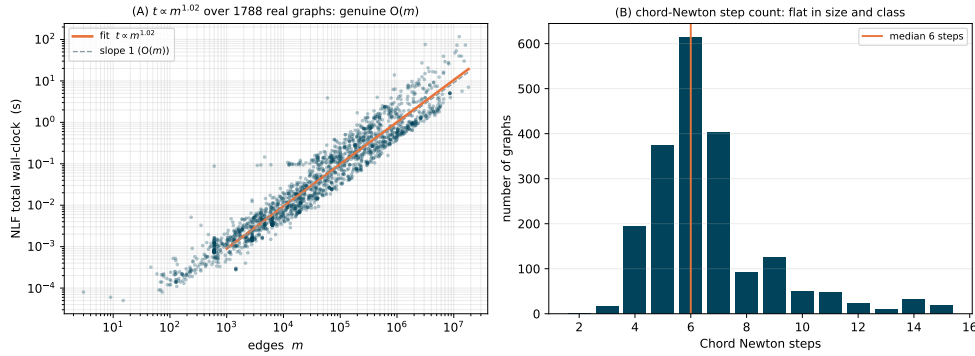


FIG. 4.2. NLF over the full real-world SuiteSparse corpus (2,003 graphs, all classes, $10^2 - 1.8 \times 10^7$ edges, 100% converged). (A) total wall-clock vs. m (log-log): the fit is $t \propto m^{1.02}$ over five decades (slope-1 reference dashed), empirically near-linear, per-edge cost flat at ≈ 0.86 μ s/edge median. (B) Newton step count: a flat distribution (median 6, range 2–15).

$n \leq 300,000$ (the subset where Ipopt’s sparse-direct KKT does not run out of memory on poorly-separable instances), each competitor capped at $5\times$ NLF’s wall-clock (Table 4.3). NLF converged on all 1,669; first-order failed on 481 (ill-conditioning, $\kappa \sim N$) and interior-point on 143 (fill-in). On the 90 graphs hard on both axes, neither competitor finished within budget. Where they converge, NLF is median $2.6\times/4.2\times$ faster than Ipopt/L-BFGS and fastest-or-only on 85%. Timing protocol: all three solvers share one global precompilation pass, and NLF is then timed on a warm per-graph call; the competitors on their first per-graph call. This per-graph warm-up favours NLF by a small-graph-weighted margin that is negligible at scale, where NLF’s advantage is the genuine avoidance of direct fill-in (§4.4); the “within budget” counts use the $5\times$ -NLF wall-clock cap and are a wall-clock criterion, not a claim the others cannot converge given more time.

TABLE 4.3

Three-paradigm robustness frontier across the SuiteSparse corpus (1,669 graphs, undirected BPR congestion—the same convex program for all). NLF (multigrid-Newton) vs. Ipopt (interior-point/sparse-direct KKT) vs. L-BFGS-on-the-dual (matrix-free first-order); each competitor capped at $5\times$ the measured NLF wall-clock (1 s floor, 120 s ceiling). NLF converges on every graph and is the only paradigm that stays within budget on both axes. On 90 graphs neither competitor finishes within the $5\times$ NLF budget, and Ipopt is additionally too large to run on 2 more—a union of 92 on which NLF alone finishes (“within budget” is a wall-clock criterion, not a claim the others cannot converge given more time).

regime			graphs	outcome
both competitors converge (well-conditioned)			1,137	NLF $\leq$ both (median $2.6\times/4.2\times$ faster than Ipopt/L-BFGS)
ill-conditioned (first-order degrades)			389	L-BFGS exceeds $5\times$ NLF; NLF, Ipopt converge
poorly-separable (direct fills in)			51	Ipopt over budget (fill-in); NLF, L-BFGS converge
both over budget $\Rightarrow$ union			92	NLF is the only solver that finishes
total			1,669	NLF 100% converged; fastest-or-only on 85%

4.7. Inner-solver ablation: which near-linear engine. The framework can use any near-linear inner solver (§3.1); we settle the default empirically. Holding NLF’s outer method fixed (load-continuation chord-Newton, lazy refresh), we swap only the inner Laplacian solve and rerun the congestion corpus of §4.5 (one BPR instance each; over 1,800 graphs except the very largest graphs). The candidates are the two public $\mathcal{O}(m)$ solvers of §3.1, approximate Cholesky and LAMG+; we exclude sparse-direct, which is not near-linear—its separability crossover against an $\mathcal{O}(m)$ inner is the scaling result of §4.4, not an inner we would ship.

Both converge on every graph, to the identical equilibrium: the dual energy agrees to floating-point precision and the link flows to $\leq 2\times 10^{-8}$ relative error. So robustness is therefore not a discriminator. Speed splits by graph class, both ways: approximate Cholesky is faster on 81% of graphs (median $1.8\times$, up to $43\times$ —its cheap factor wins on low-treewidth mesh and finite-element graphs) and LAMG+ on the other 19% (up to $24\times$ —its hierarchy wins on the poorly-separable and structural graphs); the overall median is $1.6\times$ in approximate Cholesky’s favour, with 39% of graphs within $1.5\times$ either way (Table 4.4). We therefore adopt the faster approximate Cholesky as the default and keep LAMG+ as a robust alternative: it wins wall-clock on the poorly-separable and structural minority, and at the saturating fold its cut-respecting aggregation (§3.1) gives a structurally clean near-singular solve with no fold-aware tuning, though approximate Cholesky’s constant is often smaller even there (§5.1).

Residual tolerance criterion sensitivity is mild (Table 4.5): freezing the factor more aggressively trades a few extra outer steps for fewer setups until it over-freezes, and a looser inner forcing η costs outer steps; the stock (refresh = 0.25, $\eta = 0.05$) sits in the flat interior.

4.8. Cost as a function of the target accuracy. The dependence on the requested solution accuracy ε is logarithmic. Each inner solve to relative residual δ costs $\mathcal{O}(m \log(1/\delta))$ (a bounded inner contraction gives $\mathcal{O}(\log(1/\delta))$ inner iterations). The outer iteration is a frozen chord-Newton with inexact-Newton forcing η_k tracking the nonlinear residual $\|r_k\|$ (§3.1); its per-step cycle counts form a series dominated

TABLE 4.4

Inner-solver ablation inside NLF’s BPR congestion solve, outer method fixed: LAMG+ vs. approximate Cholesky (both $\mathcal{O}(m)$). Wall-clock seconds and outer (chord-Newton) steps, to a 10^{-7} residual; both return the identical equilibrium (energy E to the digits shown; flows to Δf relative). $t_L/t_A > 1$ means approximate Cholesky is faster. A representative cross-class subset spanning both speed directions; the corpus-wide summary is in the text and the full per-graph output in the released artifact.

graph	class	m	energy E	LAMG+ (stp)	approxChol (stp)	t_L/t_A
3elt	2D mesh	13,722	−0.8939	0.050 (16)	0.015 (21)	3.3
airfoil1	2D FEM	12,289	−0.9866	0.048 (18)	0.015 (21)	3.3
bodyy5	aniso. FEM	55,346	−0.9045	0.56 (18)	0.070 (21)	8.0
bcsstk38	structural	173,714	−0.0432	0.14 (15)	0.22 (21)	0.6
G2_circuit	circuit	288,286	−1.9134	2.63 (21)	0.58 (21)	4.5
fe_ocean	FEM mesh	409,593	−1.0609	3.84 (18)	0.96 (21)	4.0
p2p-Gnutella04	P2P	39,994	−3.1625	0.14 (24)	0.16 (25)	0.9
as-caida	AS/comm	53,381	−6.6601	0.083 (20)	0.17 (23)	0.5
web-NotreDame	web	1,090,108	−25.957	4.51 (27)	3.69 (29)	1.2

TABLE 4.5

Outer-loop parameter sensitivity (approximate-Cholesky inner): outer (chord-Newton) steps to a 10^{-9} residual on representative congestion instances. Refresh—rebuild the frozen factor when a step’s residual reduction is worse than this threshold ($\eta=0.05$ fixed): flat in the interior, a few extra steps only when over-frozen (*as-caida*). Forcing η —the inner relative tolerance (refresh=0.25 fixed): a looser inner trades directly into more outer steps. The stock (refresh, η)=(0.25, 0.05) (*bold*) sits in the flat interior and is near time-optimal.

graph	refresh ($\eta=0.05$)				forcing η (refresh=0.25)		
	0.10	0.25	0.50	0.80	0.01	0.05	0.20
airfoil1	18	18	18	18	14	18	33
as-caida	21	20	29	29	17	19	25
bodyy5	18	21	18	19	15	18	36
delaunay_n16	18	18	21	18	15	18	33

by the final solve (where $\|r_k\| \rightarrow \varepsilon$), the earlier looser steps adding only a bounded geometric tail, while the continuation and globalization run at a fixed, ε -independent path tolerance. Each contribution is thus a bounded multiple of a single linear solve, so the total cost to accuracy ε is, empirically,

$$(4.2) \quad \mathcal{O}(m \log(1/\varepsilon)).$$

The constant is the whole-nonlinear-solve-to-one-linear-solve ratio reported below (2–4), and it stays bounded because the frozen chord-Newton contraction stays bounded away from 1: on the no-fold congestion class directly (§4), and through the fold because the continuation deflates the diverging cut direction and keeps the bordered corrector uniformly conditioned (§3.3); empirically the contraction does not degrade at F^* . We verify this by sweeping ε from 10^{-2} to 10^{-12} on one graph per class (web, social, road, finite-element): the total inner-solve count grows *linearly* in $\log(1/\varepsilon)$, at 0.24–0.60 per decade of accuracy, confirming (4.2).

The nonlinear/linear cost ratio is $\mathcal{O}(1)$. Brandt’s yardstick says: solving the nonlinear problem should cost only a small, size-independent multiple of solving one linear problem of the same kind [35, §8.3]. We measure exactly this: the full congestion solve (load continuation, all steps, all cycles, lazy refresh) against a single inner setup-and-solve of the linearization at $\alpha = 0$ (the linear resistor network $B \operatorname{diag}(\rho'(0)) B^\top \phi =$

d) on the same graph. Across the classes the ratio is 1.9–4.4—grid 3.1, finite-element 2.5, road 1.9, social 3.8, web (1.9×10^6 edges) 4.4—flat in size and class.

4.9. How accurately to solve each corrector step at the fold. At the fold, continuation is mandatory (J singular at F^*), but each arclength corrector need only stay on the solution path [35, §8.3.2]—a loose per-step tolerance reaches the same min cut in $2\text{--}7\times$ fewer corrector solves and hierarchy rebuilds, while a single corrector step per arclength step overshoots F^* .

5. Maximum flow. The saturating instance (2.5) recovers the exact combinatorial maximum flow. Read $f = \rho(B^\top \phi)$ as an edge-flow vector: ρ 's range is the capacity box $|f_e| \leq c_e$, and $Bf = \alpha d$ is conservation, so any ϕ solving (1.1) is a capacity-feasible, conserved flow of value α . The recession slope of Ψ_e is c_e , so E has a finite minimizer iff $\alpha < F^*$; as $\alpha \uparrow F^*$ the min-cut edges saturate ($\rho'_e \rightarrow 0$ there only) and ϕ develops a cut-indicator step. By max-flow/min-cut duality the largest feasible α is F^* , the min-cut capacity, and ϕ is the LP dual. Because $f = \rho(B^\top \phi)$ is a *nonlinear* function of the gradient, it realizes the true max-flow; a *linear* resistor network computes only the sub-maximal electrical flow (0.02–0.20 F^* on heterogeneous capacities, except single-bottleneck instances where the two coincide). The nonlinearity is essential.

Cut-respecting coarsening. As $\alpha \uparrow F^*$ the linearization J becomes singular along a single near-null direction—the min-cut indicator—so $\kappa(J) \rightarrow \infty$. We assume a *simple fold*: the min cut is unique, so the limiting near-null space is one-dimensional (a degenerate cut of multiplicity k , e.g. in symmetric unweighted graphs, would make it k -dimensional and require a rank- k block border in place of the scalar one below). The continuation of §3.3 *deflates* that one direction to a scalar outside the engine (the bordered/pseudo-arclength system); what the inner solver sees is the deflated complement (Eq. 3.4). *The deflation is the enabling property*—it is what keeps the inner operator well-conditioned through the fold, and it is a property of the framework, not of any one inner solver (swapping multigrid for a sparse-direct or another near-linear Laplacian solve reaches the identical F^* ; §5.1). Multigrid's relaxation-based affinity is a natural fit on the complement—since the forming cut is the zero-conductance set of J (§2), the affinity declines to aggregate across it, so the coarse hierarchy mirrors the cut without being told where it is—but it is helpful, not required. The measured multigrid factor *on the deflated operator* is $\mu \approx 0.01$, uniformly in α/F^* down to $0.999 F^*$ (Fig. 5.1); the uniformity is a property of the deflation, not of aggregation alone—on the undeflated J , μ degrades with $\kappa(J)$ as expected.

Exactness and scaling. NLF returns the exact max-flow (Table 5.1), checked against F^* from a linear program over free flows and from push-relabel (which agree to 10^{-4}), on every graph, to solver tolerance. The pseudo-arclength step count is flat in size (Fig. 5.1B, mean 6.7) and the total wall-clock is empirically $\mathcal{O}(m)$: $t \propto m^{1.00}$ over the max-flow sweep (to $\sim 10^4$ edges, Fig. 5.1A). *Calibration, not a contest:* Boykov–Kolmogorov, on its target vision grids, runs at $\approx 0.17 \mu\text{s}/\text{edge}$ (Fig. 5.1A), far below NLF's constant—**NLF is not meant to be a faster combinatorial max-flow solver**. Its value is being $\mathcal{O}(m)$, returning the cut potential (LP dual), the full flow-vs-load curve, and a smooth interior flow—the objects an outer method manipulates when max-flow is an inner step—one instance of a framework that, on the congestion class, has *no* combinatorial competitor at all.

5.1. Inner solver ablation. The deflation, not the choice of inner solver, is what carries the continuation through the fold (§5); the inner Laplacian solve is there-

TABLE 5.1

NLF recovers the true max-flow exactly; the gradient/electrical relaxation falls far short. F^* : LP over free flows and push-relabel agree to 10^{-4} ; *washington* and *genrmf* are DIMACS-challenge generators [22].

graph	n	F^*	grad/ F^*	NLF/ F^*
grid2d/16	256	4.973	0.20	1.0000
grid3d/6	216	17.478	0.11	1.0000
washington	66	500.745	0.02	1.0000
genrmf	64	16.000	0.00	1.0000
bottleneck	40	1.000	1.00	1.0000

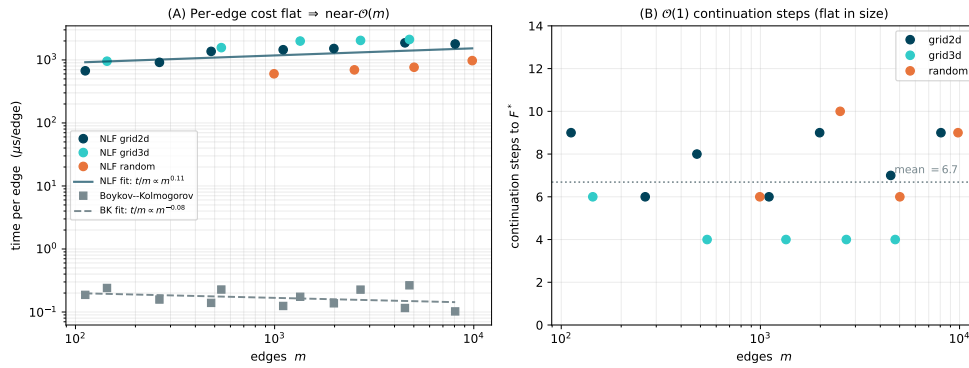


FIG. 5.1. Max-flow scaling (whole algorithm, total wall-clock), on 2D/3D grids and random graphs to $\sim 10^4$ edges, all exact at F^* . (A) Per-edge cost: nearly flat ($t \propto m^{1.00}$, empirically $\mathcal{O}(m)$) at ≈ 1.4 ms/edge, the constant set by the per-step hierarchy rebuild; Boykov-Kolmogorov sits far below at ≈ 0.17 μ s/edge (calibration only). (B) The continuation is $\mathcal{O}(1)$: pseudo-arclength steps are flat in size (mean 6.7).

fore a replaceable module. We check this directly: holding NLF’s outer method fixed (chord-Newton with the deflated pseudo-arclength continuation to F^*), we swap only the inner solver—multigrid (LAMG+), approximate Cholesky [25], an exact sparse Cholesky factorization, and a diagonally preconditioned CG—on max-flow instances spanning FEM, structural, and circuit graphs (Table 5.2).

The result is modularity, not a winner: every inner solver that converges returns the identical F^* to solver tolerance, so the continuation is genuinely solver-agnostic. The distinction that matters is near-linear versus not: the exact factorization is fastest where it fits but exhausts memory past ~ 150 k poorly-separable nodes (*G2_circuit*, *scircuit*), exactly the regime where a near-linear inner solve is mandatory, while the diagonal-PCG baseline—no coarse correction—is typically up to two orders of magnitude slower and the least scalable. Among the near-linear solvers, approximate Cholesky and LAMG+ are interchangeable in robustness; approximate Cholesky often carries the smaller constant. The single stiff case (*bcsstk38*) is instructive: there *all* iterative inner solves stall and only the exact factorization reaches F^* —the deflation removes the *global* cut singularity but not the residual ill-conditioning of the deflated system, for which an exact inner solve is still required.

TABLE 5.2

Inner-solver swap inside NLF’s max-flow continuation to F^* (wall-clock seconds; “stall” = did not converge within 120 steps; “OOM” = exact factorization size-guarded at 150k nodes). Every converging solver returns the identical F^* , so the continuation is solver-agnostic; the operative choice is near-linear versus not. Representative subset; the bake-off script and its outputs are in the released artifact. On `bcsstk38` the stalled iterative solvers were arrested at 37.0, far from the true $F^* = 45.6$ the direct solve reaches.

graph	class	n	F^*	LAMG+	approxChol	diagPCG	direct
<code>grid2</code>	2D FEM	3,296	3.003	11.4	0.75	12.3	0.24
<code>airfoil1</code>	2D FEM	4,253	6.695	29.3	1.30	20.6	0.40
<code>bodyy5</code>	aniso. FEM	18,589	3.325	307	9.5	214	2.0
<code>bcsstk38</code>	structural	8,032	45.61	stall	stall	stall	4.0
<code>circuit_4</code>	circuit	67,029	1.381	7.3	5.5	124	2.9
<code>G2_circuit</code>	circuit	150,102	4.030	1579	50.1	3011	OOM

6. Multicommodity congestion. §4 established the formulation and the $\mathcal{O}(m)$ solver for a single commodity. This section extends both, end to end, to K simultaneously routed commodities: a vector edge law that couples the commodities through shared congestion (§6.1), a solver that reuses Algorithm 3.1 and one scalar inner setup (factor or hierarchy) for all K commodities (§6.2), validation against an exact-Newton baseline (full $Kn \times Kn$ block solve, §6.3), and the same full-corpus robustness sweep as §4.5 (§6.5). The implementation is a modular layer over the single-commodity solver, which remains intact.

6.1. Formulation: a vector edge law. K commodities ship demands $\alpha d^1, \dots, \alpha d^K$ ($\mathbf{1}^\top d^k = 0$) over one network; commodity k has flow $f^k \in \mathbb{R}^m$ and potential $\phi^k \in \mathbb{R}^n$. Collect the per-edge flows into the vector $\mathbf{f}_e = (f_e^1, \dots, f_e^K) \in \mathbb{R}^K$ and the per-edge potential gradients into $\mathbf{g}_e = ((B^\top \phi^1)_e, \dots, (B^\top \phi^K)_e)$. The modeling question is how the commodities couple through shared congestion. Three natural choices: the signed total $\sum_k f_e^k$ (degenerate—collapses to single-commodity); the volume $\sum_k |f_e^k|$ (classical but nonsmooth, forces a per-edge complementarity akin to the max-flow cut); and the Euclidean magnitude $\|\mathbf{f}_e\|$ (smooth, strictly convex, reduces exactly to $K = 1$). We take the third for this feasibility study; the volume-coupled and directed models remain future work (§7.1). The program is (2.3) with the congestion cost charged to the magnitude of the commodity-flow vector,

$$(6.1) \quad \min_{f^1, \dots, f^K} \sum_e \Phi_e(\|\mathbf{f}_e\|) \quad \text{s.t.} \quad Bf^k = \alpha d^k, \quad k = 1, \dots, K,$$

and stationarity gives, per edge, the *vector law*

$$(6.2) \quad \mathbf{f}_e = \rho_e(\|\mathbf{g}_e\|) \frac{\mathbf{g}_e}{\|\mathbf{g}_e\|} :$$

the scalar law of (2.1) applied to the magnitude of the gradient vector and acting along it, so the K gradient components share one congested conductance. At $K = 1$, (6.2) is (2.1). The system $Bf^k = \alpha d^k$ with (6.2) is the stationarity of the convex energy $E(\phi^1, \dots, \phi^K) = \sum_e \Psi_e(\|\mathbf{g}_e\|) - \alpha \sum_k (d^k)^\top \phi^k$, strictly convex modulo the K per-commodity constants, so the equilibrium is unique. Physically the coupling turns on exactly where congestion does. On real capacity data at rush-hour load, commodity flows superpose to within solver tolerance ($\lesssim 10^{-4}$ relative)—real networks place each edge in one of BPR’s two power-like regimes where path splits are

load-independent—while the shared load inflates marginal costs by up to $8.8\times$: the congestion externality is slowdown, not rerouting. Synthetic instances with random capacities do reroute (12–19% flow deviation), so the corpus sweep (§6.5) exercises the solver where commodities genuinely interact.

The Newton linearization of (6.2) is the K -commodity block Laplacian whose (k, l) block is $B \text{diag}(c_e \delta_{kl} + (\rho'_e - c_e) u_e^k u_e^l) B^\top$: per edge, the $K \times K$ block

$$(6.3) \quad \frac{\partial \mathbf{f}_e}{\partial \mathbf{g}_e} = c_e I_K + (\rho'_e - c_e) \mathbf{u}_e \mathbf{u}_e^\top, \quad c_e = \frac{\rho_e(s_e)}{s_e}, \quad \mathbf{u}_e = \frac{\mathbf{g}_e}{s_e}, \quad s_e = \|\mathbf{g}_e\|,$$

with eigenvalues c_e (multiplicity $K - 1$, across the flow direction) and ρ'_e (along it): symmetric positive definite for any monotone law, with anisotropy ratio $c_e/\rho'_e = t'_e(F)F/t_e(F) \in [1, p]$ bounded by the BPR exponent.

6.2. Algorithm: one scalar hierarchy carries all K commodities. Algorithm 3.1 carries over with three substitutions and no new machinery. (i) *Stacked state.* The iterate is $\Phi = (\phi^1, \dots, \phi^K)$, the residual $R = (r^1, \dots, r^K)$, $r^k = \alpha d^k - B f^k$, and the stopping test is on the stacked norm. (ii) *A scalar frozen operator.* Rather than freeze the $K \times K$ block linearization, NLF freezes a scalar hierarchy H on the single conductance $w_e = \sqrt{c_e \rho'_e}$ —the geometric mean of the block’s two eigenvalues—and computes the correction as K independent inner solves, δ^k from H against r^k . Thus one $\mathcal{O}(m)$ inner setup serves all K commodities and all chord steps. Because BPR cost depends only on total edge flow, the $K \times K$ coupling per edge is rank-one, with eigenvalues c_e (multiplicity $K - 1$) and ρ'_e ; the w_e -preconditioned per-edge block then has eigenvalues $\{\sqrt{c_e/\rho'_e}, \sqrt{\rho'_e/c_e}\}$ —symmetric about 1 with ratio $c_e/\rho'_e \leq p$ independent of K , m , and the iterate. The per-edge conditioning is thus uniformly controlled; the realized contraction of the assembled iteration is governed by the graph-wide anisotropy and is measured at run time by the refresh monitor below, not bounded a priori. (iii) *A self-calibrating refresh monitor.* The block-diagonal frozen operator has an intrinsic rate floor set by the anisotropy that no rebuild can mitigate, so a refresh monitor measures the contraction on the step following each rebuild and treats only degradation beyond that baseline as staleness; otherwise a heavily congested instance would rebuild every step to no effect.

The cost accounting is the point of the construction. A chord step costs *one* vector-law evaluation— m scalar inversions, the same count as a single-commodity step, since only $\|\mathbf{g}_e\|$ is inverted—plus K cycles and K residual products: $\mathcal{O}(Km)$, on top of one shared $\mathcal{O}(m)$ setup. The entire K -dependence is K right-hand sides through one hierarchy.

6.3. Validation. The frozen block Jacobian (6.3) matches the finite-difference derivative of the residual; at $K = 1$ both the multigrid and the direct paths reproduce the single-commodity solver’s equilibrium, and both agree with the exact-Newton baseline (full $Kn \times Kn$ block solve); identical commodities receive identical potentials and a negated demand a negated potential (neither symmetry is imposed); and the joint-vs-alone coupling check above. Table 6.1 shows the reference comparison on real TNTF road networks at $K = 4$ (four well-separated demand dipoles, benchmark load $\alpha = 0.3 \text{median}(\{c_e\}_e)$ per commodity): NLF reaches the exact-Newton equilibrium to 10^{-10} with a handful of chord steps and a *single* hierarchy setup—whereas the exact-Newton competitor must assemble and factor the full $Kn \times Kn$ block linearization at *every* step. This is the relevant high-accuracy baseline for the coupled multicommodity program (general convex solvers such as Ipopt reduce to the same

TABLE 6.1

Multicommodity validation on real road networks (TNTP), $K = 4$. “Exact Newton” assembles the full $Kn \times Kn$ block linearization and re-solves it at every damped step (the reference); NLF is the block-diagonal chord with one frozen scalar hierarchy. Deviation is the relative ℓ_2 distance between the two equilibria.

Network	n	m	Newton steps	NLF steps	Setups	Deviation
SiouxFalls	24	38	8	8	1	5×10^{-15}
Anaheim	416	634	9	16	1	5×10^{-10}
Chicago-Sketch	933	1475	9	19	1	3×10^{-10}
Winnipeg	1040	1595	6	19	1	7×10^{-11}

the same $Kn \times Kn$ block linear algebra); NLF matches its equilibrium while replacing the K -fold-larger block solve with one shared scalar $\mathcal{O}(m)$ hierarchy.

6.4. Applications across domains. We exercise the $K = 4$ solver on the multicommodity applications the formulation targets: real TNTP metropolitan road networks (topology, capacities, free-flow times, BPR parameters, and the four heaviest origin–destination pairs all from the benchmark data, jointly scaled to a loaded rush hour, peak $\|\mathbf{f}_e\|/c_e \approx 1.07$), and corpus-protocol instances from other coupled-flow domains where flows share one network—internet AS topology, a power transmission grid, national road networks. Every instance converges to 10^{-9} with a handful of chord steps and 1–3 hierarchy setups, from 38 edges to 1.5×10^6 (e.g. **roadNet-PA**, 1.5×10^6 edges, 26 steps, 2 setups).

6.5. Corpus robustness and the cost of K . The decisive test is the same full-corpus sweep as §4.5, repeated at $K = 4$: the identical 2,003 real-world SuiteSparse graphs and instance protocol, with four well-separated demand dipoles per graph, each at the benchmark load. Every one of the 2,003 graphs converges to 10^{-9} relative residual (Fig. 6.1), to $m = 1.8 \times 10^7$ edges, at a flat median of 18 chord steps (range 9–51) and a median of 1 hierarchy setup (maximum 5; more than two on only 120 of 2,003, triggered by the refresh conditions of §3.1): the shared scalar hierarchy carries the coupled four-commodity system essentially as cheaply as it carries one. The fitted total cost is $t \propto m^{1.04}$ over four decades, median 9.1 μs per edge. Joined per graph against the single-commodity sweep, the $K = 4$ solve costs a median 3.8 $\times$ the single-commodity solve (interquartile range [3.0, 5.6])—the $\approx K$ per-step work with the setup amortized, as the cost accounting of §6.2 predicts.

The cost of K is measured directly by sweeping $K \in \{1, 2, 4, 8\}$ with nested demand sets on FEM and road graphs to 1.5×10^6 edges (**olesnik0**, **srb1**, **roadNet-PA**): the chord-step count and the hierarchy-setup count are roughly constant, and wall-clock grows close to linearly in K ($K=8$ at 4.2–8.5 $\times$ the single-commodity cost), i.e., $\mathcal{O}(Km)$ in total.

7. Extensions and Future work.

7.1. Directed multicommodity assignment. §6 routes K coupled commodities under a smooth magnitude coupling with signed flows; real traffic assignment is *directed* (nonnegative arc flows, costs coupling through the directed link flow), validated against published origin–destination equilibria (Sioux Falls, Anaheim, ...) [12, 38]. Two steps separate §6 from that target: the volume coupling $\sum_k |f_e^k|$ forces a per-edge complementarity (only commodities whose potential drop matches the common congested cost may flow)—a free boundary of the same character as the max-flow

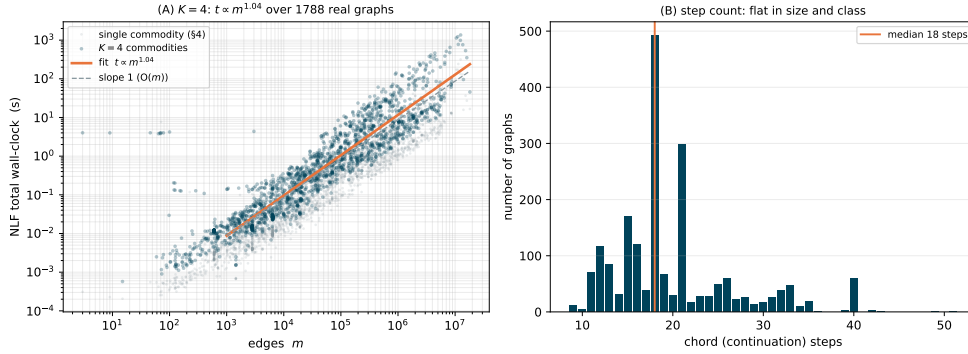


FIG. 6.1. Multicommodity NLF ($K = 4$) over the full real-world SuiteSparse corpus, mirroring the single-commodity sweep (grey, underneath). (A) Total wall-clock vs. edge count: all 2,003 graphs converge, $t \propto m^{1.04}$ over four decades. (B) Chord-step count: flat in size and class, median 18—unchanged from the single-commodity solver.

cut, attacked by the continuation-with-safeguard machinery of §3.3; and the scale of K (thousands of demand columns), made conceivable by the $\mathcal{O}(Km)$ per-step cost with one shared hierarchy (§6.2) plus per-origin aggregation borrowed from the bush-method literature [38, 39], making the K scaling conceivable. Benchmarking against Frank–Wolfe and Algorithm B on the published equilibria would be required.

The payoff: design optimization. The forward equilibrium is the means, not the end. Its motivating problems—congestion pricing, origin–destination matrix calibration, capacity network design—are bilevel programs optimized over a design x (tolls, demand, capacities) subject to the equilibrium $f(x)$, where the load-bearing object is the sensitivity $\partial f / \partial x$. Because f is the stationarity of (1.1), that sensitivity is a single *adjoint* solve with the *same* symmetric Laplacian Jacobian: the gradient of an outer objective with respect to all k design variables costs one additional $\mathcal{O}(m)$ solve, independent of k . Equilibrium sensitivity is itself classical—the adjoint/implicit-function calculus of Tobin and Friesz [42] and the bilevel network-design literature it seeds compute exactly this derivative—so the contribution is not the adjoint but that here it *is* the same $\mathcal{O}(m)$ Laplacian solve: available natively, and near-linear rather than a direct KKT factorization on poorly-separable designs. A directed single-commodity prototype we developed validates this on optimal tolling: the adjoint gradient matches finite differences. The undirected of this paper is thus a stepping stone: the same $\mathcal{O}(m)$ Laplacian machinery that *solves* (1.1) also *differentiates* it, which is exactly what bilevel network design needs.

7.2. One-shot FAS. The methodologically pure alternative to Algorithm 3.1 is a Full Approximation Scheme (FAS) multilevel cycle (local linearization) [34, 35]. In initial experiments, the LAMG+ caliber-1 interpolation misrepresented the flux feeding the coarse τ correction: caliber-1 (piecewise-constant) prolongation implicitly assumes the fine-level conductances ρ'_e are near-uniform within an aggregate, but near saturation ρ' varies by orders of magnitude across a single aggregate, so the Galerkin coarse operator $P^\top J P$ no longer represents the fine-level flux balance and the nonlinear coarse correction is inconsistent. Building a variationally faithful transfer for the saturating law—e.g. a flux-based interpolation [35, §4.6]—remains an open question.

7.3. Full multigrid: α update at the coarsest level. Brandt’s nested-iteration ideal [35, §5.6] would update the global scalar α at the coarsest level rather than the finest, prolongating (ϕ, α) upward with per-level arclength finishing. A first version was $1.6\text{--}2.1\times$ faster on small instances but $2.2\times$ slower on a larger grid (per-level finishing dominates)—a non-uniform gain—so we keep the simple $\mathcal{O}(m)$ finest-level architecture and leave a tuned FMG variant to future work.

7.4. Optimal transport as a further instance. The same edge-separable energy expresses optimal transport in Beckmann (flow) form [9]: minimizing a monotone edge-separable cost subject to $Bf = \mu - \nu$ is exactly (1.1). The W_p Wasserstein distance [40] between measures μ and ν on a graph corresponds to the ℓ^p edge cost; the quadratic case ($p = 2$, H^{-1} cost) reduces to a single $\mathcal{O}(m)$ Laplacian solve, while the W_1 limit ($p \rightarrow 1$) gives the non-smooth saturating member, a natural fold instance. The quadratically-regularized form, whose dual Hessian is an active-subgraph Laplacian solved in the literature by direct factorization [41], is where a near-linear inner could help; whether it is competitive is an open question, since the W_1 /Beckmann limit is a min-cost flow with strong combinatorial and direct solvers.

7.5. DC optimal power flow. The DC approximation to optimal power flow (DC-OPF) minimizes generation cost subject to power balance and thermal line limits:

$$(7.1) \quad \min_{\theta, p^g} \sum_i C_i(p_i^g) \quad \text{s.t.} \quad B \text{diag}(b) B^\top \theta = p^g - p^d, \\ |b_e(B^\top \theta)_e| \leq c_e, \quad p^{\min} \leq p^g \leq p^{\max},$$

where b_e is the line susceptance, p^d fixed demand, and C_i a convex (typically quadratic or piecewise-linear) generation cost. The power-balance constraint is a susceptance-weighted graph Laplacian; the line limits are the box constraints of the saturating law (2.5), so (7.1) is an instance of (1.1) extended by a nodal cost and generator box constraints.

We merely record DC-OPF as an instance of the framework but do not solve the constrained problem here: the generator box constraints and nodal cost $\sum_i C_i(p_i^g)$ require an FAS extension [34] that carries p^g and its bounds into the coarse levels (§7.2), which we have not built. As a back-of-envelope projection only—not a measured result—current IPM solvers (MATPOWER, PowerModels.jl) run 20–50 Newton steps, each a fresh sparse-direct KKT factorization; on near-planar grids (118 to 30,000 buses) we timed that factorization at $2\text{--}6\times$ cheaper than one NLF cycle, so whether a continuation-Newton loop reusing one frozen hierarchy would win on total cost turns on the iteration-count trade and is an open question.

8. Conclusion. Placing the nonlinearity in the *edge law* collapses a family of convex, edge-separable network-flow equilibria—electrical, congestion, minimum-delay, maximum-flow—onto one nonlinear Laplacian $B\rho(B^\top \phi) = \alpha d$, solved by a damped chord-Newton on a frozen linearization cycled against the nonlinear residual. Any near-linear Laplacian solver is shown to empirically translate into the same complexity on the whole nonlinear class.

Reproducibility. Every table is produced by released scripts, on public benchmarks—the SuiteSparse collection (each instance reduced to its largest connected component, §4.5), the Transportation Networks road data [12], and MATPOWER grids [2]—and on seeded synthetic generators (Erdős–Rényi topologies for the scaling study) where noted. The interior-point comparator is Ipopt 3.14.19

(MUMPS 5.9.0, 120s CPU budget); the BPR law uses the strictly convex regularization of §4.1; capacity/free-flow distributions and demand seeds are fixed and recorded. The inner solvers are public [25, 8]; the NLF solver, instance-generation scripts, and a reproducibility notebook that re-runs the timings and checks them against these tables are at <https://github.com/orenlivne/nlf>. All timings were collected on a MacBook Pro with Apple M5 Pro (18-core: 6 efficiency + 12 performance) and 48 GB unified memory, running a single Julia process with no explicit parallelism.

Appendix A. Why each application is an instance of the framework.

Per application: equilibrium principle $\Leftrightarrow$ convex program (2.3) $\Leftrightarrow$ KKT $\Leftrightarrow B\rho(B^\top\phi) = \alpha d$. Each instance differs only in the edge law; directed complementarity is absorbed into that law or relaxed by the signed formulation.

The chain in general. Stationarity of (2.3) in f_e gives the nonlinear Ohm law $t_e(f_e) = (B^\top\phi)_e$; inversion gives $f_e = \rho_e((B^\top\phi)_e)$; and Kirchhoff ($Bf = \alpha d$) yields (1.1). Summing along any source–sink path telescopes: every path has marginal cost $\phi_s - \phi_t$, the scalar equilibrium cost. The signed formulation has no inequality constraints—a smooth equation—which is why Newton with a near-linear Laplacian solve applies with no active sets.

Electrical flow (linear law). Ohm’s law is linear, $f_e = g_e/r_e$; the energy is the dissipated power and (2.3) is Thomson’s principle (current minimizes dissipation) [26]. Equation (1.1) is the ordinary weighted Laplacian system—the framework’s fixed point under linearization, and the object the inner solver [8] is built for.

Congestion (traffic) equilibrium. Wardrop’s principle—per origin–destination pair, path flows $h_p \geq 0$ with $\sum_p h_p = d$ obeying $T_p - \pi \geq 0$ and $h_p(T_p - \pi) = 0$ —is, by Beckmann’s theorem [9], the KKT system of $\min_h \sum_e \int_0^{f_e} t_e$: differentiating gives $\partial F/\partial h_p = \sum_{e \in p} t_e = T_p$, and π (the multiplier of $\sum_p h_p = d$) is both the marginal cost of demand and the equilibrium travel time. In link variables the demand multiplier unfolds into the node field ϕ ($\pi = \phi_s - \phi_t$) and the program becomes (2.3) with $\Phi'_e = t_e$, the BPR latency of §4. The signed relaxation keeps the first Wardrop clause exactly (all routes at potential-drop cost π) and discards $h_p \geq 0$: an unused route may carry counterflow—the undirected scope, with per-commodity nonnegativity the free-boundary extension of §7.

Minimum-delay routing. M/M/1 queueing with Little’s law gives the per-packet link delay $t_e(f) = 1/(c_e - f)$, and Kleinrock’s independence approximation makes total delay edge-separable [15]. Taking the delay as marginal cost, $\Phi_e = \log(c_e/(c_e - f))$, gives (2.3) directly; the inverse law $\rho_e(g) = c_e - 1/g$ (odd-extended) saturates at capacity with conductance $\rho'_e = (c_e - f)^2 \rightarrow 0$, placing the problem in the fold class of Table 2.1: the capacity constraint $f < c_e$ is never imposed—the law cannot produce it—and the feasibility limit, where the filling edges form a cut in the spare capacities, is the fold handled by §3.3.

Maximum flow. The LP $\max \alpha$ s.t. $Bf = \alpha d$, $|f_e| \leq c_e$ has the box constraint as its only nonsmoothness; its complementarity (an edge is either slack with zero reduced cost or saturated on the cut) is absorbed into the smooth saturating law (2.5), whose energy $\Psi_e = c_e^2 \log \cosh(g/c_e)$ is quadratic for slack edges and exactly linear, $c_e|g|$, for saturated ones. The active set is not tracked: it *emerges* as the zero-conductance set $w_e \rightarrow 0$, the forming min cut, and the LP’s degeneracy reappears as the one scalar fold at $\alpha = F^*$ traversed by the continuation of §3.3. At the fold the saturated set is the exact min cut and ϕ steps across it (§5); no $(1 + \epsilon)$ relaxation is involved.

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